Report - Py.tatine

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Abstract

This paper presents the approach adopted by the team for an image retrieval competition, which required retrieving the top 10 most similar images to a given query image. The dataset included both real and AI-generated images of celebrities. Performance was evaluated using a weighted top-k accuracy metric, assigning 600 points for Top-1, 300 for Top-5, and 100 for Top-10 matches. During the model selection phase, the team explored two main categories of architectures: traditional convolutional neural networks (such as ResNet and EfficientNet) and transformer-based models (e.g., DINOv2 and CLIP). After extensive experimentation, several models were identified as promising candidates for fine-tuning, with the expectation of improved accuracy. Nevertheless, the best-performing model on the day of the competition was a non-fine-tuned CLIP model in its RN50x64 variant.

1. Introduction

1.1. Task Description

The machine learning competition focused on top-k similar images retrieval, a relevant computer vision task with many applications across different domains. Participants were challenged to develop models capable of identifying and ranking the 10 most similar images to a given query from a large database. The core objective was to retrieve the most similar synthetic images from a gallery when given natural query images of celebrities. The training set contained approximately 5000 images organized into subfolders by celebrity identity, i.e. with each subfolder containing both natural and synthetic images of the same individual. The test set, on the other hand, presented a more constrained scenario: query set consisted of approximately 3000 real images of celebrities, while the gallery contained only synthetic ones. This set up (which was unknown before the competition day) tested models' ability to connect authentic photographs and AI-generated synthetic representations of the same celebrities (and in most cases, the same photo too). For each query image, models were required to retrieve 10 gallery images ranked by similarity, with the most similar image ranked first. The challenge, in fact, laid not onyl in identifying the correct celebrity but also in properly ordering the retrieved results based on visual similarity confidence.

1.2. Overview of approaches

Our team explored multiple architectural paradigms to address this cross-domain retrieval challenge, employing both traditional computer vision approaches and modern foundantion models.

- **Traditional CNN Architectures**: starting from traditional convolutional neural networks (including EfficientNet and ResNet), we implemented fine-tuning strategies. These architectures were adapted from their original image classification objectives to learn similarity embeddings suitable for retrieval tasks.
- **Transformer-based Models**: given that the details of the task were unknown until the day of the competition, the team expanded the investigation to include transformer-based architectures. We implemented DINO, a self supervised vision transformer that learns robust visual representations through knowledge distillation. Additionally, we employed CLIP (Contrastive Language-Image Pre-training), specifically the RN50x64 variant (known for the strong performance in visual tasks), which creates joint vision-language embeddings through contrastive learning on large-scale image-text pairs.

Our approach evolved from conventional supervised fine-tuning towards more sophisticated methods, recognizing the limitations of traditional CNNS for cross-domain tasks, ultimately focusing on models that were pre-trained on diverse data that could capture semantic similarity beyond pixel-level correspondance.

1.3. Summary of results

The competition revealed significant differences in performance across architectural choices. Despite extensive fine-tuning efforts on the provided training data, both EfficientNet and ResNet architectures demonstrated poor performance across all evaluation metrics. These models struggled significantly with the natural-to-synthetic domain transfer, suggesting that traditional convolutional features learned through supervised fine-tuning were insufficient for bridging the substantial visual gap between authentic photographs and AI-generated celebrity images. During the competition, CLIP RN50x64 emerged as the decisive winner, consistently and substantially outperforming all other implemented approaches across all evaluation metrics. Indeed, CLIP RN50x60 achieved a score of 734.693/1000, summing up scores in Top-1 Accuracy (600 points), Top-5 Accuracy (300 points), and Top-10 Accuracy (100 points). This superior performance can be attributed to CLIP's unique contrastive pre-training methodology, which learns to associate visual and textual representations across massive datasets, resulting in more semantically meaningful embeddings that generalize effectively across visual domains.

These results highlight a fundamental shift in computer vision capabilities, where foundation models (i.e. an AI neural network — trained on mountains of raw data, generally with unsupervised learning — that can be adapted to accomplish a broad range of tasks) trained on diverse, large-scale data with sophisticated objectives substantially outperform traditional architectures, even when the latter are specifically fine-tuned for the target domain. The findings emphasize the critical importance of pre-training methodology and semantic understanding for challenging cross-domain retrieval tasks in modern computer vision applications.

Code and model configurations presented in this paper are available at: https://github.com/brunelliMichele/py.tatine

2. Models considered

In this section, we present the set of machine learning models explored for the image retrieval competition. Given the lack of prior knowledge about the test distribution and the type of images used for the task, we adopted a diversified modeling strategy that spanned across both classical convolutional neural networks and more recent transformer-based architectures.

Each model was evaluated in terms of its ability to generate discriminative image embeddings suitable for content-based retrieval. To this end, we experimented with different levels of supervision, transfer learning paradigms (from fixed feature extraction to full and partial fine-tuning), and similarity metrics.

The following architectures were considered:

- **EfficientNet**: chosen for its balance between efficiency and accuracy, tested in frozen mode as a feature extractor.
- **ResNet**: used under three distinct settings — fixed features, full fine-tuning, and selective fine-tuning — to assess how different training strategies affect retrieval quality.
- **VGG-16**: included as a classical baseline to benchmark against more modern architectures and to validate the stability of the retrieval pipeline.
- **DINOv2**: tested as a self-supervised transformer model, capable of producing general-purpose visual embeddings without explicit fine-tuning.
- **CLIP**: the most competitive model in our experiments, evaluated across multiple variants and chosen for fine-tuning based on a trade-off between semantic generalization and computational efficiency.

For all architectures, we extracted image representations which were L2-normalized and indexed using different methods, depending on the model and setup. Cosine similarity was the default metric for comparing embeddings, unless stated otherwise. In the following sections, we describe the specific implementation details and retrieval performance considerations for each model family.

2.1. EfficientNet

EfficientNet is a convolutional neural network (CNN) architecture that introduces a systematic scaling method to optimize model performance. Instead of arbitrarily increasing network depth, width, or input resolution, EfficientNet uniformly scales these three dimensions using a compound scaling coefficient. This approach is based on the following intuitions:

- **Larger input images require greater depth** to capture long-range dependencies
- **Increased width** is needed to learn more detailed and fine-grained features
- **Higher resolution** inputs should be accompanied by proportional increases in both depth and width in order to maintain a balanced and efficient network structure

By coordinating these scaling dimensions through fixed scaling factors, EfficientNet achieves a better trade-off between accuracy and efficiency compared to traditional CNN architectures [**?**]. For the competition, we performed feature extraction using EfficientNetB0 and EfficientNetB4 models, both pretrained on ImageNet and accessed via TensorFlow's keras.applications module. To reduce the spatial dimensions of the output feature maps, we applied a Global Average Pooling (GAP) layer, which averages the value of each feature map, transforming a 3D tensor into a 1D vector of fixed length that is more suitable for similarity comparison. For the retrieval task, we used Annoy (Approximate Nearest Neighbors Oh Yeah) to perform fast and memory-efficient nearest neighbor searches. Although the initial implementations used Euclidean distance, we transitioned to cosine similarity, which better suits the nature of deep image embeddings. This was implemented by L2-normalizing the extracted feature vectors before indexing them, ensuring that angular distance (as used by Annoy) corresponds to cosine distance.

Annoy is particularly well-suited for this use case because:

- it supports **angular distance** which is well suited for normalized embeddings
- it is **highly efficient and scalable**, even with high dimensional data
- enables **fast retrieval times** with low memory usage thanks to the tree-based structure.

The attempt to fine-tune EfficientNet did not provide meaningful improvement in model's performance, which is why it was not tested on competition day.

2.2. ResNet

ResNet (Residual Networks) is a deep convolutional neural network architecture and it introduces residual (or skip) connections, which allow gradients to flow directly thorugh the network, enabling training of much deeper architecture. OOur investigation employed three distinct ResNet-based approaches for content-based image retrieval, each implementing different transfer learning paradigms and similarity computation methodologies. We utilized a pre-trained ResNet50 model as a fixed feature extractor, leveraging representations learned from ImageNet classification. The network's final fully connected layer was replaced with an identity mapping, transforming the model from a classifier to a feature encoder. This modification enabled extraction of 2048-dimensional feature vectors from the global average pooling layer preceding the original classification head. L2-normalized feature vectors were compared using cosine distance to find the 10 most similar images to the query. The second methodology employed ResNet18 architecture with full network fine-tuning using cross-entropy loss for multiclass classification on the target domain. Following training convergence, the classification head was removed, and 512-dimensional feature vectors were extracted from the penultimate layer. Similarity search was implemented using Facebook's FAISS (Facebook AI Similarity Search) library with L2 (Euclidean) distance indexing. The third strategy implemented selective fine-tuning of ResNet50, where all convolutional layers remained frozen (requires_grad=False) while only the final classification layer underwent parameter updates. This conservative transfer learning approach preserves the robust feature representations learned from ImageNet while adapting the decision boundary to the target classification task. After training, the model was suited for feature extraction (by removing the trained classification layer), leading to 2048-dimensional normalized vectors used for similarity computations using cosine distance.

2.3. VGG

VGG-16 is a classical convolutional architecture known for its simplicity and uniform design, consisting of sequential 3×3 convolutional layers followed by max-pooling and fully connected layers [**?**]. We decided to try it because of its layer-wise simplicity and consistent use of small convolutional filters make it interpretable and stable across a variety of visual tasks. We included VGG-16 as a diagnostic baseline to validate our retrieval pipeline and to assess how well a classical CNN performs on a modern cross-domain retrieval task. However, in our retrieval scenario, it exhibited several limitations. Its strong reliance on local texture patterns made it susceptible to the types of superficial variations introduced by synthetic image generation — such as lighting inconsistencies, skin smoothness artifacts, or uniform backgrounds. Additionally, the absence of skip connections limited its ability to integrate deep semantic context, which proved essential for cross-domain identity recognition. Retrieval rankings often favored images with similar lighting or pose over those with true identity matches. These results aligned with prior findings that CNNs trained on ImageNet are overly texture-biased [**?**],

2.4. DINO V2

Meta's DINOv2 [**?**] is a recent self-supervised vision transformer (ViT) trained using a teacher-student framework that avoids the need for manual labels. Unlike traditional supervised learning, DINO trains models to maximize consistency across different augmented views of the same image. This promotes the learning of features that are robust to low-level variations such as texture, lighting, and background, and encourages a focus on object-level or identity-preserving semantics. For those reasons it showcases an ability to generalize across different visual domains without the need for fine tuning, which made it an obvious choice, given the task. Specifically we used the vit_base_patch16_224.dino model. This corresponds to a Vision Transformer (ViT-B/16) trained using the original DINO self-supervised framework [**?**], due to its capacity to form semantically coherent representations without requiring human-labeled supervision. Neverthless, we obtained very low performances during the competition, probably because Dino is not trained to distinguish between identities and it lacks explicit supervision for facial or fine-grained matching tasks, as argued by Oquab et al. [**?**]. Additionally, adding a fine-tuning layer did not help; in fact, the performance decreased. Our most likely explanation for such a result is that being pre-trained with unlabeled semantic relationships, fine-tuning it on a labeled training set might distort its geometry [**?**].

2.5. CLIP

The CLIP (Contrastive Language–Image Pretraining) framework, developed by OpenAI, was the most effective model we evaluated. CLIP is trained on hundreds of millions of image-text pairs scraped from the internet, learning to align images and natural language descriptions in a shared embedding space. This multimodal contrastive training equips CLIP with a robust understanding of high-level semantics, making it especially well-suited for retrieval tasks where visual similarity is more conceptual [?]. We tested several variants of CLIP in frozen mode, including ViT-B/16, ViT-L/14, and RN50x64. All models were evaluated by computing cosine similarity between image embeddings. Among them, RN50x64 obtained the best performance during the competition window. Another factor in the choice to deploy it was that it is trained on a wide range of image-text pairs during training, CLIP generalizes well to new domains and unseen classes and it should also achieve high performances in unlabeled settings. As seen, CLIP models come in multiple architectural forms — most notably Vision Transformers (ViTs) and ResNet-based (RN) backbones. We chose to evaluate both because each brings different inductive biases and practical trade-offs that can influence retrieval performance depending on the task. While ViT based models are particularly adept at capturing long-range semantic relationships and context-aware representations [?], ResNet based models are often more robust to small-scale distortions, faster to train or fine-tune, and more efficient in memory-constrained environments. Their convolutional structure also tends to preserve fine-grained spatial details that are useful in identity-level retrieval [?]. We decided to test both because of the unknown nature of the retrieval task we would face. During competition, we choose to fine-tune only RN50x64 for time issues, and we choose it because we hypotized that ResNets would be more stable during fine-tuning, especially on small datasets; also RN50x64 is significantly more memory-efficient than ViT-L/14, which was key given the context.

3. Evaluation

This section presents a detailed evaluation of the models developed by our team. Consistent with the structure of the competition, the evaluation is divided into two main phases: pre-competition and post-competition assessment.

Since the official competition dataset and evaluation metric were not available beforehand, we relied on the ”Garden of Eden” dataset (available on Kaggle[1]), which consists of labeled images of flowers organized into 104 class-specific subfolders. We used the training portion of the dataset to build our gallery set, as the folder structure provided clear ground-truth labels for each image.

To simulate retrieval queries, we randomly selected a subset of images from the dataset's test split. Each query image was assigned its corresponding class label based on its folder of origin. For each query, we performed a top-10 retrieval from the gallery set using our model. The evaluation was performed by checking the folder (i.e., class label) of each retrieved image. If a retrieved image came from the same folder as the query image, it was considered a correct match. Otherwise, it was counted as incorrect. Precision@K was then computed as the proportion of correct matches among the top-k retrieved results.

[1] https://www.kaggle.com/datasets/msheriey/104-flowers-garden-of-eden

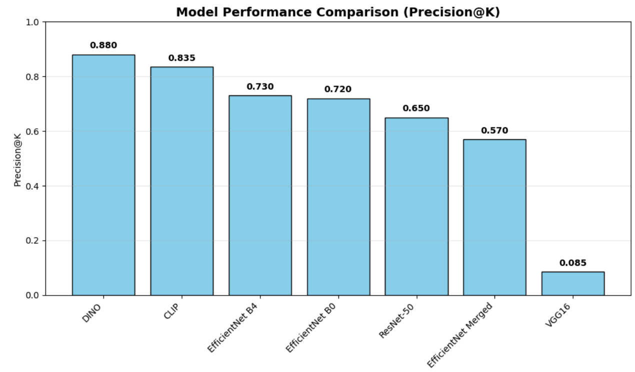

Figure 1. Comparison of model performance using Precision@K on the flower dataset.

In contrast, the post-competition evaluation was conducted using the official competition dataset and a custom metric designed specifically for the image retrieval task: weighted top-accuracy. This metric not only accounts for whether a relevant image appears in the top-k results but also assigns scores based on its rank. Specifically:

- 600 points are awarded if the correct image is ranked first (top-1 accuracy),
- 300 points if it appears within the top-5,
- and 100 points if it is among the top-10.

This ranking-sensitive evaluation incentivizes models to prioritize the most relevant matches at the top of the retrieval list.

3.1. Pre-competition evaluation

The pre-competition evaluation aimed to identify the most promising candidates among several pre-trained models using the Kaggle dataset. Our objective was to determine the optimal hyperparameter configurations for each model to enhance their performance. This strategy allowed us to optimize time management during the competition. We compared models based on performance metrics, specifically Precision@K and top-k accuracy, as well as their computational efficiency. Based on these comparisons, we selected a lineup of models to focus on during the competition. This approach enabled us to pinpoint the best-performing pre-trained and fine-tuned models for the unknown task in advance, allowing us to concentrate efforts on refining its results on competition day.

The pre-competition evaluation revealed significant performance variation across different achitectural approaches, as shown in Fig. **??**.

These results reveal a marked performance gap between self-supervised models such as DINO and CLIP and traditional convolutional neural networks (CNNs). Both DINO and CLIP significantly outperformed traditional CNNs, achieving Precision@K scores above 0.8, with DINO emerging as the top performer.

DINO's Vision Transformer architecture excels at capturing semantic similarities between flower images. Thanks to its self-supervised learning approach, it effectively learns rich visual representations without requiring explicit labels, resulting in outstanding retrieval performance where semantic understanding is critical.

CLIP, while slightly trailing DINO in precision, demonstrated strong performance despite not being explicitly optimized for single-domain retrieval tasks like ours. This aligns with CLIP's known robustness in extracting generalizable and discriminative features across varied domains.

Traditional CNN architectures showed moderate but limited performance. Within the EfficientNet family, the larger and more complex B4 variant offered only marginal gains over the more compact B0 version (0.73 vs. 0.72 Precision@K), indicating that architectural enhancements in EfficientNet-B4 provide limited benefit over B0 for this specific task. ResNet-50 exhibited moderate pre-trained performance (0.65), yet we opted to fine-tune it for several key reasons:

- **Proven Architecture**: ResNet-50 is a widely established and extensively studied CNN with a deep architecture known for strong generalization when adapted to new domains.
- **Modularity and Compatibility**: Its modular design and broad framework support make it an ideal candidate for fine-tuning, offering a favorable balance between complexity and training stability.
- **Potential for Improvement**: Given the lack of prior domain-specific information for the retrieval task, we hypothesized that fine-tuning ResNet-50 could substantially enhance its retrieval capabilities.
- **Practical Evaluation**: Fine-tuning a moderately performing model like ResNet-50 allowed us to measure the gains achievable through domain adaptation starting from a non-optimal baseline. This was crucial for assessing the robustness and flexibility of our overall pipeline.

Conversely, VGG16 performed poorly (Precision@K of 0.085), likely due to its relatively shallow architecture and limited representational capacity compared to more modern networks. Its simpler feature extraction proved insufficient for capturing the nuanced visual similarities necessary for effective retrieval.

Having witnessed the exceptional performance of CLIP and DINO during pre-competition evaluation, we chose to prioritize them for deeper investigation. In particular, CLIP stood out because of its multiple variants, each offering a different trade-off between accuracy and computational requirements. After benchmarking popular CLIP backbones—such as ViT-B/32, ViT-L/14, and RN50x64—we selected the CLIP RN50x64 variant. This model provided the best balance of high retrieval accuracy and moderate computational cost, outperforming the lighter ViT-B/32 and being more resource-efficient than the memory-intensive ViT-L/14. This choice enabled efficient inference without sacrificing performance, making it well-suited for our competition constraints.

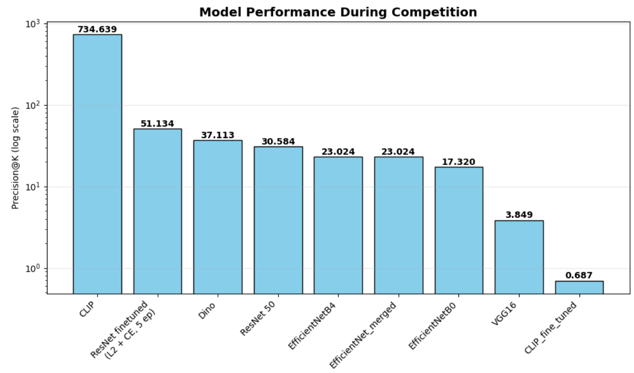

Figure 2. Comparison of model performance using weighted top-k accuracy on competition dataset.

3.2. Post competition evaluation

The post-competition evaluation was conducted using the official competition dataset and the weighted top-accuracy metric described earlier. Critically, the actual competition task differed dramatically from our pre-competition scenario: instead of retrieving similar flower images within the same domain, teams were challenged to retrieve synthetic celebrity images from a gallery when given natural query images of the same celebrities. This cross-domain retrieval scenario introduced a substantial domain shift, fundamentally altering model performance and offering valuable insights into their generalization capabilities (see Fig. **??**). Success in this task required models to disregard superficial domain-specific patterns and instead focus on identity-preserving features that remained consistent between natural and synthetic images.

CLIP emerged as the clear winner with an outstanding score of 734.64, outperforming all other models and validating our strategy to focus on this architecture during competition day. CLIP's training on image-text pairs enabled it to learn high-level semantic representations, trascending domain boundaries. Instead of being build on purely visual similarity, as the other models, CLIP's multimodal training through exposure to diverse visual representations paired with consistent semantic labels led to an outstanding performance. Moreover, the choice of the RN50x64 variant proved to be optimal for handling the complexity of the cross domain face recognition task, and some technical implementation incread its potentiality: the feature extraction pipeline included L2 normalization and cosine similarity as distance measure. These choices proved ideal for identity-based matching, where the directional relationship between features matters more than their magnitude, as in cross-domain retrieval tasks absolute features may vary significantly between natural and synthetic images.

In contrast, although DINO performed strongly in the pre-competition phase, the natural-to-synthetic celebrity matching task exposed its limitations in cross-domain settings. DINO is pre-trained on natural images from the ImageNet dataset, making it effective at capturing visual similarity rather than person-specific identity. This explains its success on the ”Garden of Eden” dataset, a homogeneous set of images where both query and gallery images share the same visual domain. Moreover, in our implementation of the Vision Transformer-based model (DINO), we extracted the image representation using the CLS token, a special vector generated by the model that is designed to summarize the entire input image. Specifically, we used the output at index 0 of the transformer sequence, which corresponds to this classification token. While the CLS token is effective at capturing global semantic information, it is not explicitly trained to focus on fine-grained, identity-specific features such as facial structure or distinguishing characteristics. As a result, the model likely encoded more generic visual information, such as background textures, lighting variations, and other contextual elements, which tend to differ significantly between natural and synthetic images. This limited its ability to isolate the consistent, identity-preserving features necessary for successful cross-domain celebrity matching.

Traditional CNN architectures exhibited relatively poor performance, with their limitations becoming especially evident in the cross-domain celebrity retrieval task. Among the EfficientNet variants, both the B4 and the merged versions achieved comparable scores (23.02), outperforming the smaller B0 variant (17.32). VGG16 performed the worst, with a Precision@K of only 3.85.

However, fine-tuning ResNet-50 led to a significant improvement, boosting its score from 30.58 (baseline) to 51.13 (a 67% increase) demonstrating that domain adaptation can partially mitigate the challenges posed by cross-domain data when sufficient training samples from both domains are available.

Nonetheless, while domain-specific fine-tuning can enhance the retrieval capabilities of CNNs, the relatively modest gains suggest that traditional CNN architectures face inherent limitations compared to modern self-supervised models, exhibiting reduced adaptability to domain shifts and producing less semantically rich feature representations.

Table 1. Model Comparison for Image Retrieval. Fine-tuned models present epochs, other models are pre-trained. Pooling is reported only where explicitly modified in the code.

Model	Batch	Epochs	LR	Pooling	Dist.	Loss	Top@k
VGG16	16	–	–	–	Cosine	–	3.85
CLIP RN50x64	16	–	–	–	Cosine	–	734.63
DINOv2	–	–	–	CLS token	Cosine	–	37.11
EffNet B0	–	–	–	GAP	Euclidean	–	17.31
EffNet B4	–	–	–	GAP	Angular	–	23.02
EffNet B4 (M)	–	–	–	GAP	Angular	–	23.02
ResNet18 (FT)	32	5	1e-4	–	–	CrossEntropy	51.13
ResNet50 (FT)	32	20	0.001	–	–	CrossEntropy	53.15
ResNet50 Ret.	16	–	–	–	Cosine	–	30.58

4. Final Considerations

This report presented a comprehensive investigation of different machine learning architectures, ranging from traditional CNNs to modern foundation and transformer-based models, for cross-domain image retrieval. The most significant finding was the great superiority of CLIP RN50x64, which achieved a weighted top-k accuracy score of 734.63 out of 1000 points, substantially outperforming all other evaluated approaches. This performance gap was far from marginal, demonstrating a fundamental shift in how semantic understanding translates to retrieval effectiveness. Traditional CNN architectures, even when fine-tuned on domain-specific data, struggled to bridge the natural-to-synthetic domain gap. CLIP RN50x64, in fact, is built on a wide ResNet-50 backbone, with 64 channels per layer - significantly wider than standard ResNet-50. Moreover, its architecture integrates contrastive learning to align image and text embeddings in a shared semantic space, enabling robust multimodal representations that generalize well across domains. While traditional CNNs excel at capturing visual patterns, CLIP's multimodal training enables it to learn concepts on a more abstract level, thereby transcending superficial visual differences, without the need for fine-tuning on specific datasets, as demonstrated by Iijima *et al.* [**?**]. This capability proved essential for recognizing that a natural photograph and its AI-generated synthetic version represent the same individual, despite substantial visual differences. Our use of the ”Garden of Eden” flower dataset for pre-competition evaluation, which lacked the domain mismatch present in the actual competition, limited its usefulness as a predictor of cross-domain generalization performance, thus providing meaningful insights into the aforementioned capabilities. The notable contrast in performance between DINO (top performer on flowers, but heavy underperformer on celebrities) and CLIP (top performer on celebrities) underscores how training methodology plays a more crucial role than architectural complexity or robustness. This finding highlights that self-supervised learning on natural images (DINO) excels within domains but struggles with cross-domain generalization, while contrastive multimodal training (CLIP) develops more robust semantic representations. DINO, despite using a transformer-based architecture known for global reasoning [**?**], lacks the explicit semantic grounding provided by paired supervision, limiting its effectiveness on identity-preserving tasks. That implies, more in depth, that the poor performance of DINO in the setting of the competition can be attributed to a combination of factors. As a self-supervised model trained solely on natural images without explicit labels or image-text alignment, DINO learns to group images based on appearance-level consistency rather than semantic identity. This makes it particularly effective for clustering similar textures, colors, or shapes, but not necessarily for associating different domains of the same concept, such as a real and synthetic portrait of the same person. Additionally, DINO's representations tend to focus on mid-level or instance-agnostic patterns [**?**], which are insufficient for high-level semantic matching across domains. In practice, DINO struggles to extract identity-preserving features, resulting in a retrieval strategy that favors visually similar but semantically unrelated images. Furthermore, the cross-domain nature of the competition task exposed several limitations in traditional computer vision approaches. Our results demonstrate that pixel-level similarity, as typically captured by CNN features, is insufficient for bridging significant domain gaps. The synthetic images likely exhibited different lighting, textures, and rendering features that ”confused” models trained primarily on natural image statistics. CLIP's exceptional performance illustrates the critical distinction between visual similarity and semantic similarity. Ultimately, CLIP's success suggests that exposure to diverse image-text pairs during pretraining encourages the development of semantically meaningful, identity-preserving features, which are less sensitive to domain-specific variations. This makes CLIP RN50x64 especially suitable for tasks involving cross-modal or cross-domain matching.

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